

IT3021
Data Warehousing & Business Intelligence

Assignment 2

Submitted By: Karunartahne U.M.T.N

IT23679726

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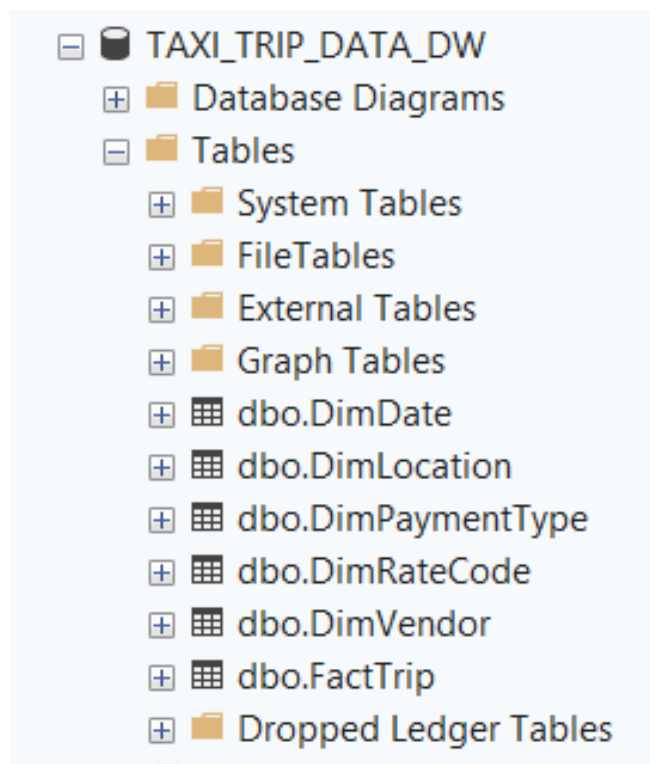
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1. Data source for the assignment.

This assignment utilizes the data warehouse developed in Assignment 1 as the primary data source. The data warehouse is designed based on the NYC Taxi Trip dataset and follows a star schema architecture to support analytical processing and reporting.

The central fact table in the warehouse is the FactTrip table, which stores transactional data for each taxi trip. Each record in the FactTrip table represents a single taxi trip and contains measurable attributes such as trip distance, fare amount, total amount, passenger count, and trip duration.

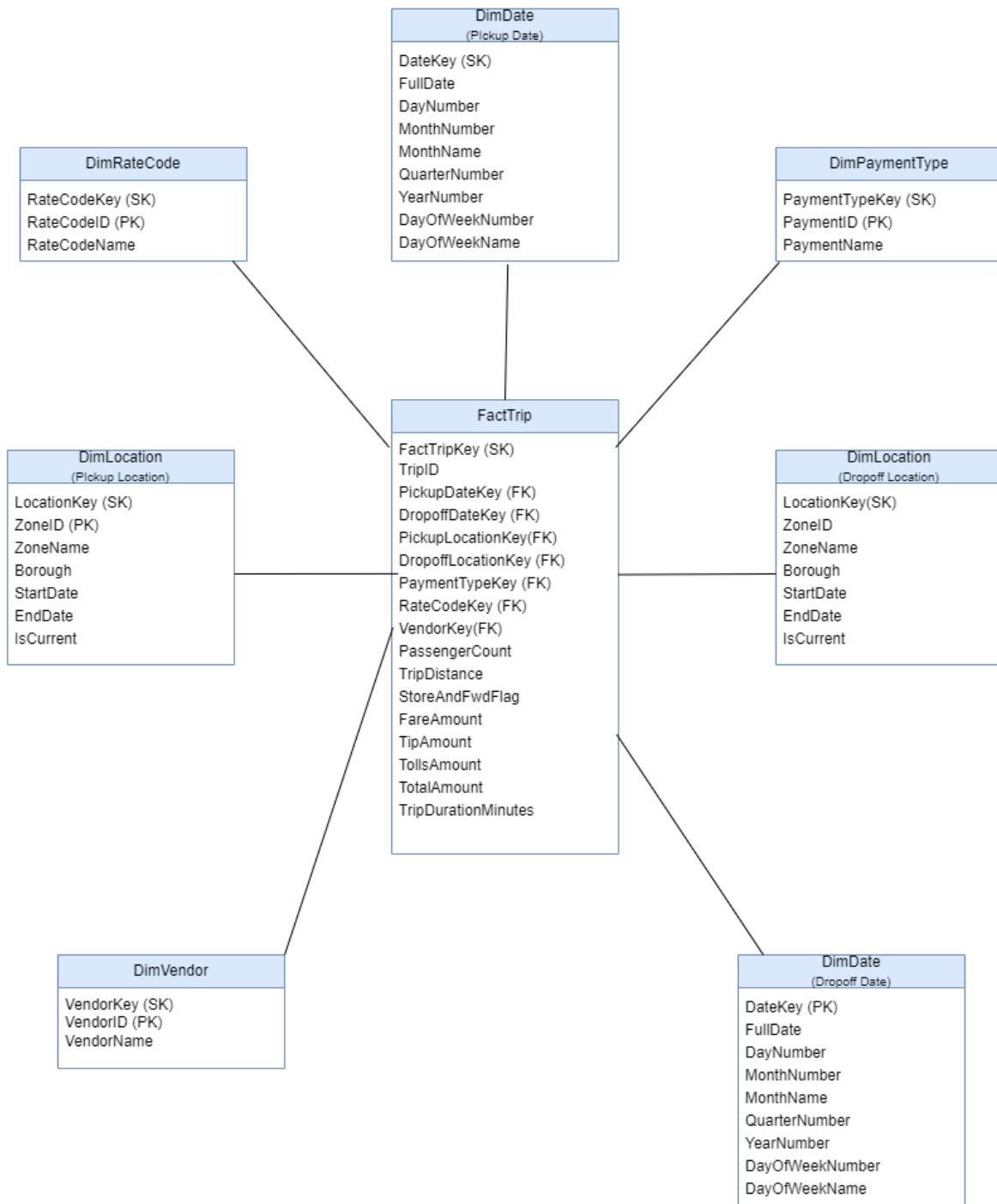
The data warehouse also implements Slowly Changing Dimensions (SCD), particularly in the DimLocation table, to track changes in location-related attributes over time.

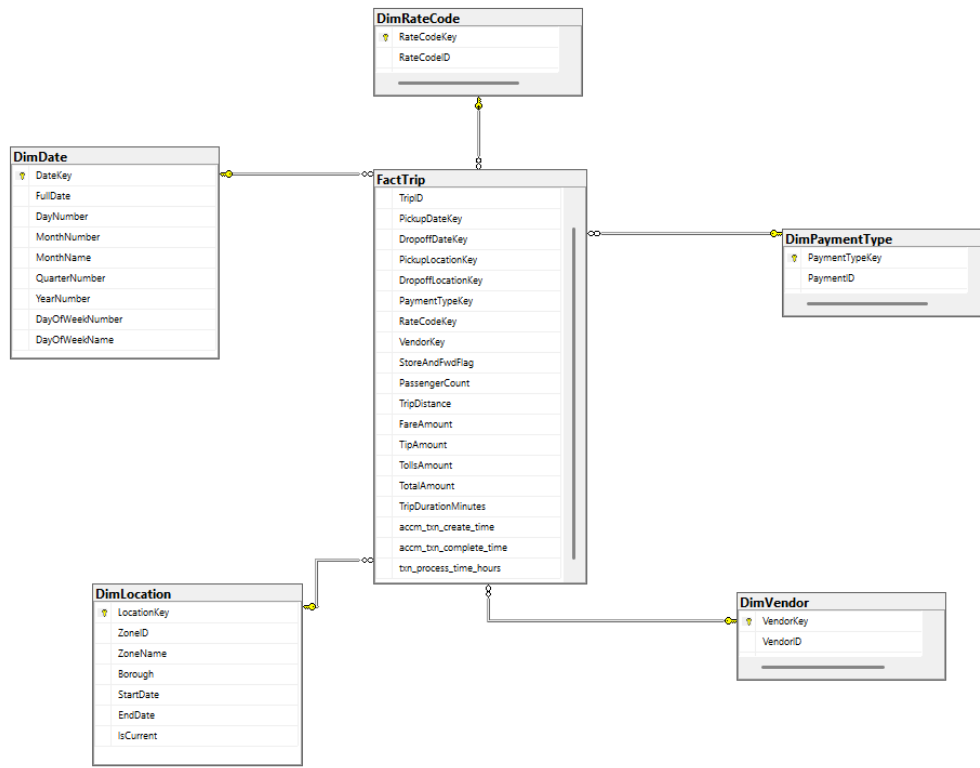


MSI\MSSQLS...aymentType

Column Name	Data Type	Allow Nulls
PaymentTypeKey	int	☐
PaymentID	int	☐
PaymentName	varchar(50)	☐
		☐

Column Properties





100 % No issues found Ln: 11, Ch: 1 (34 chars, 1 lines) TABS CRLF UTF-8

Results	Messages	FactTripKey	TripID	PickupDateKey	DropoffDateKey	PickupLocationKey	DropoffLocationKey	PaymentTypeKey	RateCodeKey	VendorKey	StoreAndFwdFlag	PassengerCount	TripDistance	FareAmount	TipAmount	Tolls
1		1	TRIP_1	20180329	20180329	481	267	25	27	12	N	1	18.15	70.00	16.16	10.5
2		2	TRIP_2	20180329	20180329	443	494	25	25	12	N	1	4.59	25.00	5.16	0.00
3		3	TRIP_3	20180329	20180329	495	495	25	25	12	N	1	0.30	3.00	0.76	0.00
4		4	TRIP_4	20180329	20180329	495	349	25	25	12	N	2	16.97	49.50	5.61	5.76
5		5	TRIP_5	20180329	20180329	455	349	25	25	12	N	5	14.45	45.50	10.41	5.76
6		6	TRIP_6	20180329	20180329	451	349	25	25	12	N	1	11.60	42.00	14.57	5.76
7		7	TRIP_7	20180329	20180329	458	455	25	25	11	N	1	5.80	24.00	4.95	0.00
8		8	TRIP_8	20180329	20180329	475	481	25	25	12	N	1	3.38	25.00	5.16	0.00
9		9	TRIP_9	20180329	20180329	455	267	25	27	12	N	1	16.98	85.00	15.00	12.5
10		10	TRIP_10	20180329	20180329	443	481	25	25	12	N	1	4.99	22.00	4.76	0.00

100 % 

 No issues found

 Results

 Messages

	LocationKey	ZoneID	ZoneName	Borough	StartDate	EndDate	IsCurrent
1	267	1	Newark Airport	EWB	2026-03-26	NULL	1
2	268	3	Allerton/Pelham Gardens	Bronx	2026-03-26	NULL	1
3	269	18	Bedford Park	Bronx	2026-03-26	NULL	1
4	270	20	Belmont	Bronx	2026-03-26	NULL	1
5	271	31	Bronx Park	Bronx	2026-03-26	NULL	1
6	272	32	Bronxdale	Bronx	2026-03-26	NULL	1
7	273	46	City Island	Bronx	2026-03-26	NULL	1
8	274	47	Claremont/Bathgate	Bronx	2026-03-26	NULL	1
9	275	51	Co-Op City	Bronx	2026-03-26	NULL	1
10	276	58	Country Club	Bronx	2026-03-26	NULL	1

	VendorKey	VendorID	VendorName
1	11	1	Creative Mobile Technologies, LLC
2	12	2	VeriFone Inc.

100 %		No issues found	
Results		Messages	
	PaymentTypeKey	PaymentID	PaymentName
1	25	1	Credit Card
2	26	2	Cash
3	27	3	No Charge
4	28	4	Dispute
5	29	5	Unknown
6	30	6	Voided Trip

	RateCodeKey	RateCodeID	RateCodeName
1	25	1	Standard Rate
2	26	2	JFK
3	27	3	Newark
4	28	4	Nassau or Westchester
5	29	5	Negotiated Fare
6	30	6	Group Ride

2. SSAS Cube implementation

2.1. Create the Data Source

Select how to define the connection
You can select from a number of ways in which your data source will define its

☐ Create a data source based on another object

☒ Create a data source based on an existing or new connection

Data connections:

Property	Value
Data Source	MSI\MSSQLSERVER2
Initial Catalog	TAXI_TRIP_DATA_DW
Persist Securit...	True
Provider	SQLNCLI11.1
User ID	sa

New... Delete

< Back Next > Finish >>| Cancel

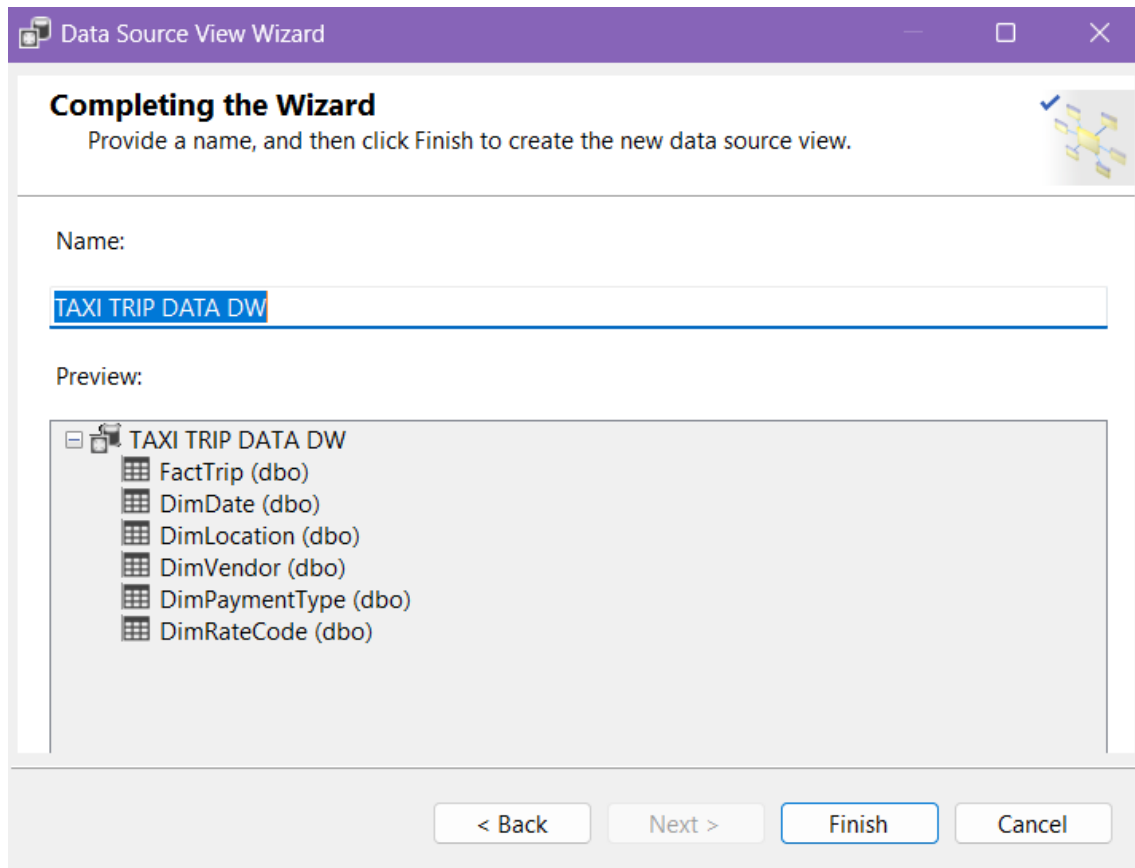
Completing the Wizard
Provide a name and then click Finish to create the new data source.

Data source name:
TAXI TRIP DATA DW

Preview:

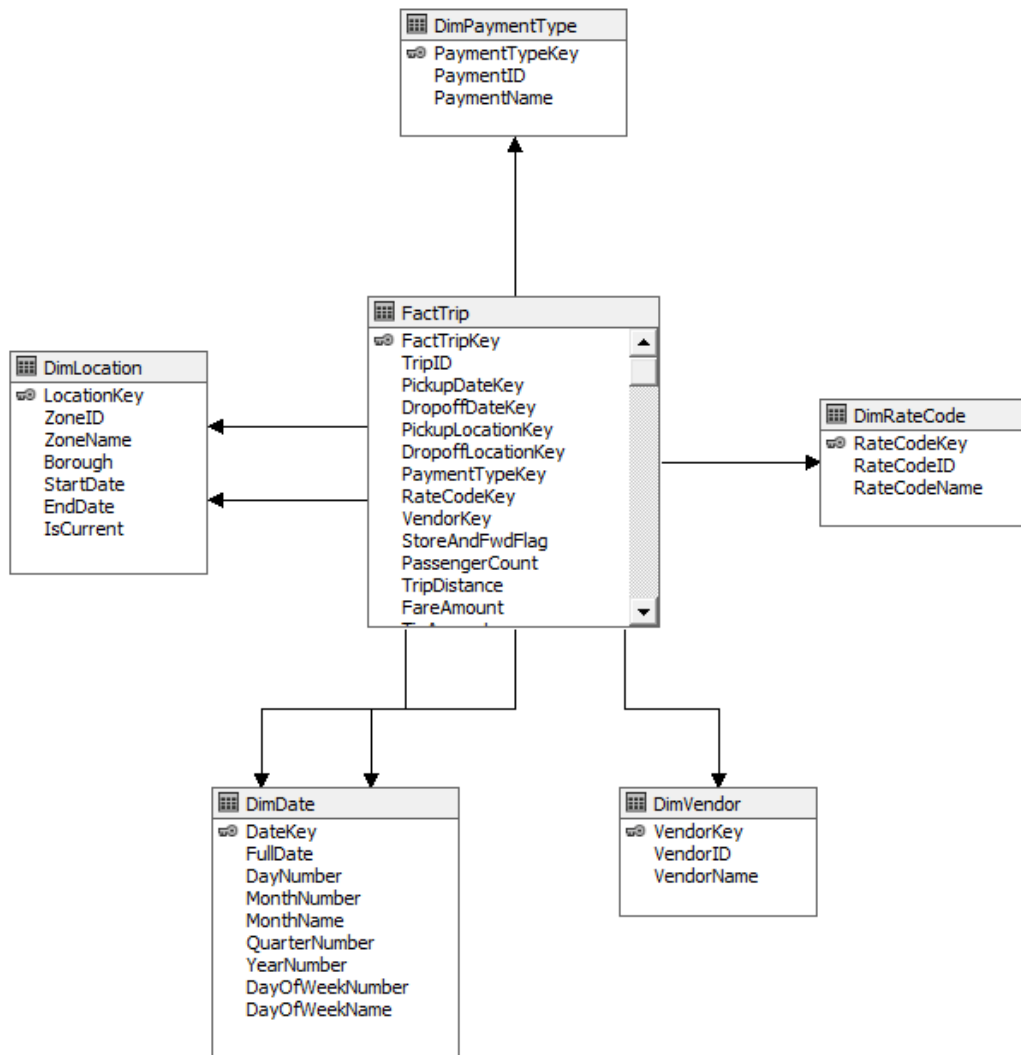
Connection string:
Provider=SQLNCLI11.1;Data Source=MSI\MSSQLSERVER2;Persist Security Info=True;Password=;User ID=sa;Initial Catalog=TAXI_TRIP_DATA_DW

< Back Next > Finish Cancel



2.2.Create the Data Source View

A Data Source View (DSV) was created to define the logical model used by the cube. The DSV displayed the relationships between the fact table and the dimension tables and served as the basis for dimension and cube creation.



2.3.Create Dimensions.

Dimension Wizard

Select Dimension Attributes
Specify dimension attributes and select Enable Browsing to surface them as hierarchies.

Available attributes:

☒ Attribute Name	☒ Enable Browsing	Attribute Type
☒ Date Key	☒	Regular
☒ Full Date	☒	Regular
☒ Day Number	☒	Regular
☒ Month Number	☒	Regular
☒ Month Name	☒	Regular
☒ Quarter Number	☒	Regular
☒ Year Number	☒	Regular
☒ Day Of Week Number	☒	Regular
☒ Day Of Week Name	☒	Regular

< Back Next > Finish >>| Cancel

Dimension Wizard

Completing the Wizard
Type a name for the new dimension, verify the dimension structure, and then click Finish to save the dimension.

Name:
Dim Date

Preview:

- Dim Date
 - Attributes
 - Date Key
 - Full Date
 - Day Number
 - Month Number
 - Month Name
 - Quarter Number
 - Year Number
 - Day Of Week Number
 - Day Of Week Name

< Back Next > Finish Cancel

Dimension Wizard

Select Dimension Attributes

Specify dimension attributes and select Enable Browsing to surface them as hierarchies.

Available attributes:

☒	Attribute Name	☒ Enable Browsing	Attribute Type
☒	Location Key	☒	Regular
☒	Zone ID	☒	Regular
☒	Zone Name	☒	Regular
☒	Borough	☒	Regular
☐	Start Date	☐	Regular
☐	End Date	☐	Regular
☐	Is Current	☐	Regular

< Back
Next >
Finish >>|
Cancel

Dimension Wizard

Completing the Wizard

Type a name for the new dimension, verify the dimension structure, and then click Finish to save the dimension.

Name:

Preview:

Dim Location

- Attributes
 - Location Key
 - Zone ID
 - Zone Name
 - Borough

< Back
Next >
Finish
Cancel

 Dimension Wizard

Select Dimension Attributes

Specify dimension attributes and select Enable Browsing to surface them as hierarchies.



Available attributes:

☒	Attribute Name	☒ Enable Browsing	Attribute Type
☒	Payment Type Key	☒	Regular
☒	Payment ID	☒	Regular
☒	Payment Name	☒	Regular

< Back

Next >

Finish >>|

Cancel

 Dimension Wizard

Completing the Wizard

Type a name for the new dimension, verify the dimension structure, and then click Finish to save the dimension.



Name:

Preview:

 Dim Payment Type

 Attributes

 Payment Type Key
  Payment ID
  Payment Name

< Back

Next >

Finish

Cancel

Dimension Wizard

Completing the Wizard

Type a name for the new dimension, verify the dimension structure, and then click Finish to save the dimension.

Name:

Dim Rate Code

Preview:

Dim Rate Code

Attributes

Rate Code Key

Rate Code ID

Rate Code Name

< Back

Next >

Finish

Cancel

Dimension Wizard

Select Dimension Attributes

Specify dimension attributes and select Enable Browsing to surface them as hierarchies.

Available attributes:

☒ Attribute Name	☒ Enable Browsing	Attribute Type
☒ Rate Code Key	☒	Regular
☒ Rate Code ID	☒	Regular
☒ Rate Code Name	☒	Regular

< Back

Next >

Finish >>|

Cancel

Dimension Wizard

Completing the Wizard

Type a name for the new dimension, verify the dimension structure, and then click Finish to save the dimension.

Dim Vendor

Preview:

Dim Vendor

Attributes

Vendor Key

Vendor ID

Vendor Name

< Back

Next >

Finish

Cancel

Dimension Wizard

Select Dimension Attributes

Specify dimension attributes and select Enable Browsing to surface them as hierarchies.

Available attributes:

☒	Attribute Name	☒ Enable Browsing	Attribute Type
☒	Vendor Key	☒	Regular
☒	Vendor ID	☒	Regular
☒	Vendor Name	☒	Regular

< Back

Next >

Finish >>

Cancel

2.4.Create Hierarchies

User-defined hierarchies were created to improve cube navigation and support drill-down and roll-up analysis.

Calender Hierarchy – Year, Quarter, Month, FullDate

Location Hierarchy – Borough, Zone Name

The screenshot displays the 'Attribute Relationships' tab in a BI tool. The top navigation bar includes 'Dimension Struct...', 'Attribute Relationships' (selected), 'Translations', and 'Browser'. Below the navigation bar, a hierarchy diagram shows 'Date Key' (with a checkmark icon) connected to 'Full Date', 'Month Name', 'Quarter Number', and 'Year Number'. The main workspace is divided into two panes: 'Attributes' on the left and 'Attribute Relationships' on the right. The 'Attributes' pane lists various date-related attributes: Date Key, Day Number, Day Of Week Name, Day Of Week Number, Full Date, Month Name, Month Number, Quarter Number, and Year Number. The 'Attribute Relationships' pane shows a list of attributes with their corresponding dimension keys: Date Key (Day Number, Day Of Week Name, Day Of Week Num..., Full Date, Month Number, Month Name, Quarter Number, Year Number). Below the main workspace, there is a second hierarchy diagram showing 'Location Key' (with a checkmark icon) connected to 'Borough' (also with a checkmark icon). The bottom pane shows the 'Attributes' list for the location hierarchy: Borough, Location Key, Zone ID, and Zone Name. The 'Attribute Relationships' pane for this hierarchy shows: Borough (Zone Name), Location K... (Borough), and Location K... (Zone ID).

Attributes	Attribute Relationships
Date Key	Day Number
Day Number	Day Of Week Name
Day Of Week Name	Day Of Week Num...
Day Of Week Number	Full Date
Full Date	Month Number
Month Name	Month Name
Month Number	Quarter Number
Quarter Number	Year Number
Year Number	

Attributes	Attribute Relationships
Borough	Zone Name
Location Key	Borough
Zone ID	Zone ID
Zone Name	

Hierarchies

Calendar Hierarchy

- Year Number
- Quarter Number
- Month Name
- Full Date
- <new level>

To create a new hierarchy, drag an attribute here.

Hierarchies

Location Hierarchy

- Borough
- Zone Name
- <new level>

To create a new hierarchy, drag an attribute here.

2.5.Create the Cube.

A multidimensional cube named TaxiTripCube was created using FactTrip as the measure group table.

Cube Wizard

Select Measure Group Tables

Select a data source view or diagram and then select the tables that will be used for measure groups.

Data source view:

TAXI TRIP DATA DW

Measure group tables:

☒ FactTrip
 ☐ DimDate
 ☐ DimLocation
 ☐ DimVendor
 ☐ DimPaymentType
 ☐ DimRateCode

Suggest

< Back

Next >

Finish >>|

Cancel

Cube Wizard

Select Measures

Select measures that you want to include in the cube.

☒ Measure

- ☒ **Fact Trip**
- ☒ Passenger Count
- ☒ Trip Distance
- ☒ Fare Amount
- ☒ Tip Amount
- ☒ Tolls Amount
- ☒ Total Amount
- ☒ Trip Duration Minutes
- ☒ Txn Process Time Hours
- ☒ Fact Trip Count

< Back Next > Finish >>| Cancel

Cube Wizard

Select Existing Dimensions

Select existing dimensions to include in the cube.

☒ Dimension

- ☒ Dim Date
- ☒ Dim Location
- ☒ Dim Vendor
- ☒ Dim Payment Type
- ☒ Dim Rate Code

< Back Next > Finish >>| Cancel

Cube Wizard

Completing the Wizard

Name the cube, review its structure, and then click Finish to save the cube.

Cube name:

Preview:

Measure groups

Fact Trip

- Passenger Count
- Trip Distance
- Fare Amount
- Tip Amount
- Tolls Amount
- Total Amount
- Trip Duration Minutes
- Txn Process Time Hours
- Fact Trip Count

Dimensions

- Dim Date
- Dim Location
- Dim Vendor
- Dim Payment Type
- Dim Rate Code

< Back

Next >

Finish

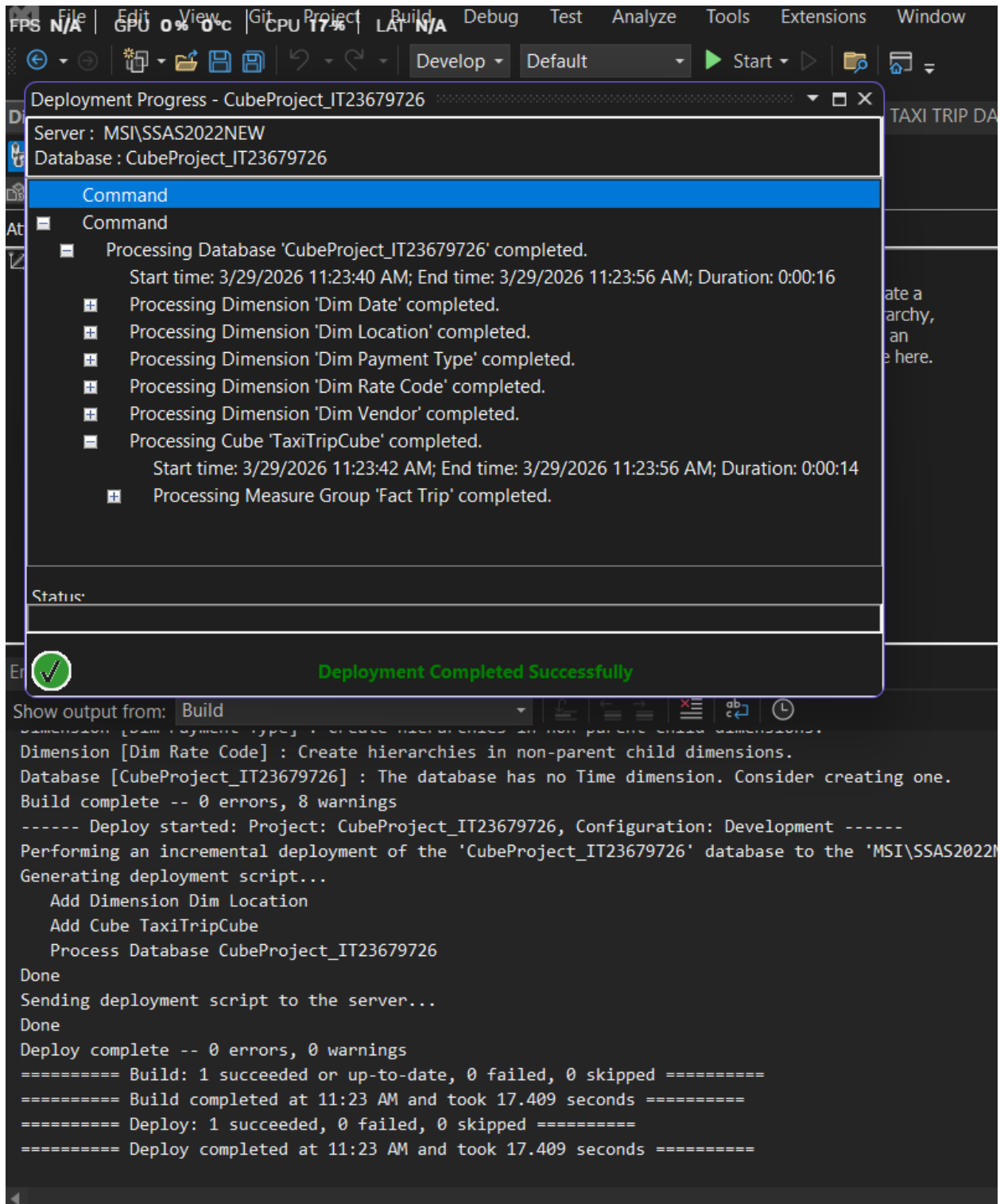
Cancel

Measures	
	TaxiTripCube
	Fact Trip
	Passenger Count
	Trip Distance
	Fare Amount
	Tip Amount
	Tolls Amount
	Total Amount
	Trip Duration Minutes
	Txn Process Time Hours
	Fact Trip Count
Dimensions	
	TaxiTripCube
	Pickup Date
	Dim Rate Code
	Dropoff Location
	Pickup Location
	Dropoff Date
	Dim Vendor
	Dim Payment Type

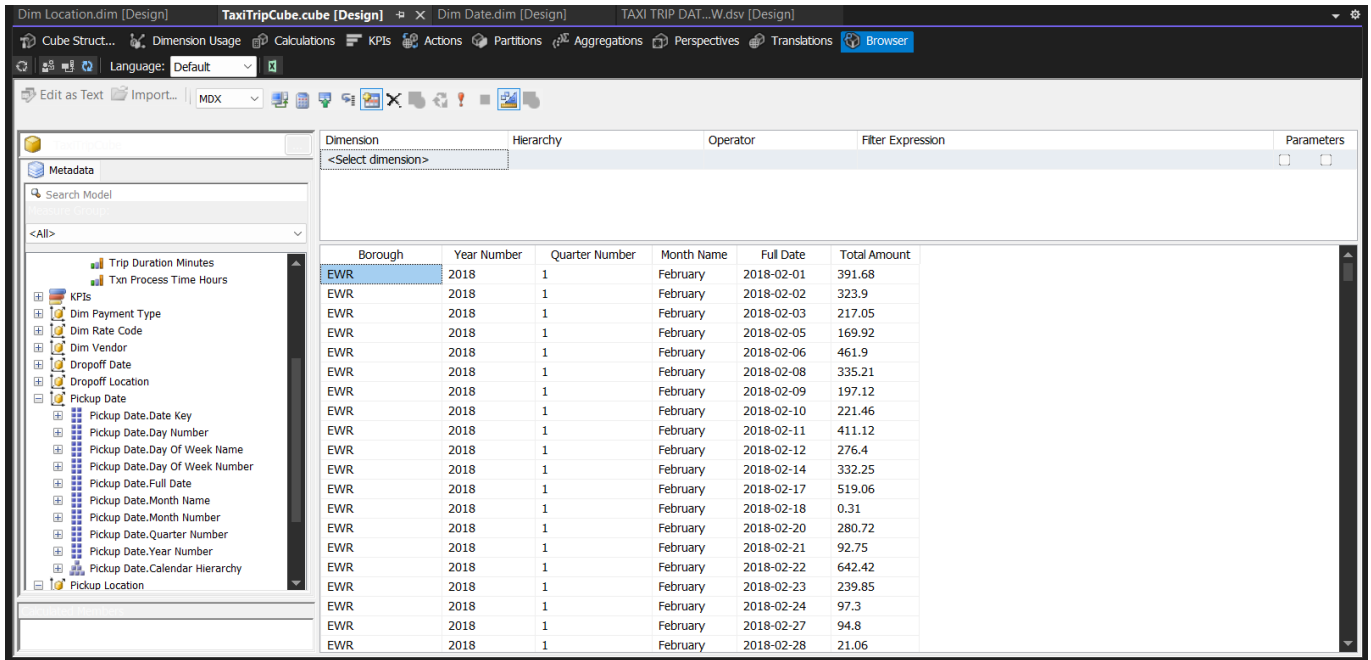
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2.6. Deploy and Process the Cube

The cube was deployed to the Analysis Services server and processed successfully.



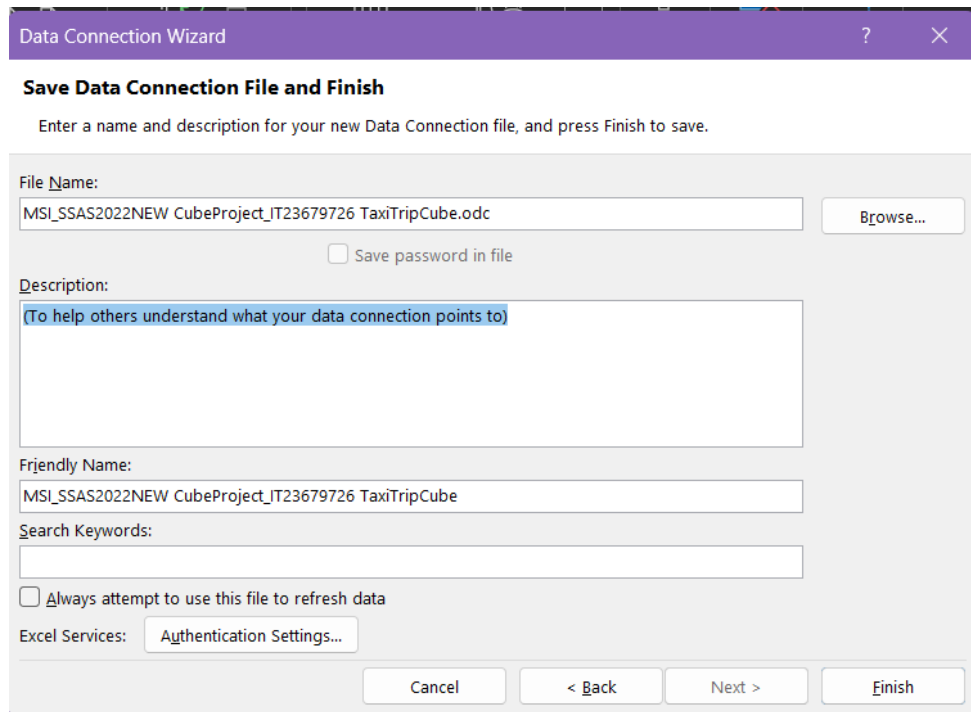
2.7. Validate the Cube in Browser



Dimension	Hierarchy	Operator	Filter Expression	Parameters
<Select dimension>				
Borough	Year Number	Quarter Number	Month Name	Full Date
EWR	2018	1	February	2018-02-01
EWR	2018	1	February	2018-02-02
EWR	2018	1	February	2018-02-03
EWR	2018	1	February	2018-02-05
EWR	2018	1	February	2018-02-06
EWR	2018	1	February	2018-02-08
EWR	2018	1	February	2018-02-09
EWR	2018	1	February	2018-02-10
EWR	2018	1	February	2018-02-11
EWR	2018	1	February	2018-02-12
EWR	2018	1	February	2018-02-14
EWR	2018	1	February	2018-02-17
EWR	2018	1	February	2018-02-18
EWR	2018	1	February	2018-02-20
EWR	2018	1	February	2018-02-21
EWR	2018	1	February	2018-02-22
EWR	2018	1	February	2018-02-23
EWR	2018	1	February	2018-02-24
EWR	2018	1	February	2018-02-27
EWR	2018	1	February	2018-02-28

3. Excel Connection and OLAP Operations

The deployed SSAS cube was connected to Microsoft Excel using the Analysis Services connection option.



Data Connection Wizard

Save Data Connection File and Finish

Enter a name and description for your new Data Connection file, and press Finish to save.

File Name:
MSI_SSAS2022NEW CubeProject_IT23679726 TaxiTripCube.odc

☐ Save password in file

Description:
(To help others understand what your data connection points to)

Friendly Name:
MSI_SSAS2022NEW CubeProject_IT23679726 TaxiTripCube

Search Keywords:

☐ Always attempt to use this file to refresh data

Excel Services: Authentication Settings...

Cancel < Back Next > Finish

3.1. Initial PivotTable

A basic PivotTable was created to confirm that Excel was successfully connected to the cube. In this report, Total Amount was used as the measure and Payment Name was used as the row field.

The screenshot displays the Microsoft Excel interface with a PivotTable and the PivotTable Fields task pane. The PivotTable is located in the range A4:B9, with 'Row Labels' in column A and 'Total Amount' in column B. The data includes categories like Cash, Credit Card, Dispute, and No Charge, along with a Grand Total. The PivotTable Fields task pane on the right shows the configuration for the PivotTable, with 'Payment Name' selected as the row field and 'Total Amount' as the value field.

Row Labels	Total Amount
Cash	3527875.7
Credit Card	46647183.4
Dispute	20421.11
No Charge	90159.92
Grand Total	50285640.13

PivotTable Fields

Choose fields to add to report:

Search

☐ Dim Payment Type
☐ Payment ID

☒ **Payment Name**
☐ Payment Type Key

☐ Dim Rate Code

Drag fields between areas below:

Filters	Columns

Rows	Values
Payment Name	Total Amount

☐ Defer Layout Update Update

3.2.Roll-up

The roll-up operation was demonstrated using the Calendar Hierarchy in the Pickup Date dimension. Data was summarized at the Year level to provide a higher-level aggregated view of Total Amount.

The screenshot shows the Microsoft Excel interface with the PivotTable Fields task pane open. The PivotTable is located in the worksheet, and the task pane shows the 'Pickup Date.Calendar Hierarchy' expanded to 'Year Number'. The PivotTable has 'Row Labels' and 'Total Amount'.

Row Labels	Total Amount
2018	50285640.13
Grand Total	50285640.13

The PivotTable Fields task pane shows the following fields:

- Dropoff Location
 - Dropoff Location.Location Hierarchy
 - More Fields
- Pickup Date
 - ☒ Pickup Date.Calendar Hierarchy
 - Year Number

The task pane also includes sections for Drag fields between areas below:

- Filters
- Columns
- Rows
 - Pickup Date.Calendar Hierarchy
- Values
 - Total Amount

At the bottom of the task pane, there is a checkbox for 'Defer Layout Update' and an 'Update' button.

3.3.Drill-down

The drill-down operation was demonstrated by expanding the Calendar Hierarchy from Year to lower levels such as Month and Date. This provided a more detailed view of the revenue data over time.

The screenshot shows the Microsoft Excel interface with a PivotTable. The PivotTable is set to show data for the year 2018, with the calendar hierarchy expanded to the date level. The PivotTable Fields task pane on the right shows the hierarchy: Dropoff Location, Pickup Date, and Year Number. The Values field is set to Total Amount. The PivotTable data is as follows:

Row Labels	Total Amount
2018	
1	
February	
2018-02-01	10924.1
2018-02-02	1342.74
2018-02-03	84073.05
2018-02-04	82574.85
2018-02-05	110457.28
2018-02-06	110225.82
2018-02-07	104668.01
2018-02-08	143073.37
2018-02-09	167860.21
2018-02-10	124642.68
2018-02-11	122675.22
2018-02-12	141210.8
2018-02-13	148837.51
2018-02-14	135098
2018-02-15	132132.54
2018-02-16	118407.47
2018-02-17	101206
2018-02-18	107839.23
2018-02-19	121205.08
2018-02-20	129250.82
2018-02-21	153453.14
2018-02-22	145790.54
2018-02-23	133636.51
2018-02-24	86766.44
2018-02-25	91996.08
2018-02-26	102775.59
2018-02-27	121759.11
2018-02-28	160801.92
January	3514212.15
March	7985281.89
2	12777832.3
3	10615123.1

3.4.Slice

The slice operation was performed by applying a single filter to the PivotTable. For example, the report was filtered by one Vendor Name, allowing the data to be viewed for only one selected vendor.

A7					
	A	B	C	D	E
2	Vendor Name	VeriFone Inc.			
4	Row Labels	Total Amount			
5	Cash	2180451.13			
6	Credit Card	28694864.36			
7	Dispute	-8656.82			
8	No Charge	-15776.06			
9	Grand Total	30850882.61			
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3.5.Dice

The dice operation was demonstrated by applying multiple filters simultaneously. In this case, the PivotTable was filtered using Vendor Name and Rate Code Name, producing a smaller subset of cube data for analysis.

The screenshot displays the Microsoft Excel interface with a PivotTable and the PivotTable Fields task pane. The PivotTable is located in the range A1:B9 and shows the following data:

Row Labels	Total Amount
Cash	1127.72
Credit Card	7194.8
Dispute	246.6
No Charge	401.68
Grand Total	8970.8

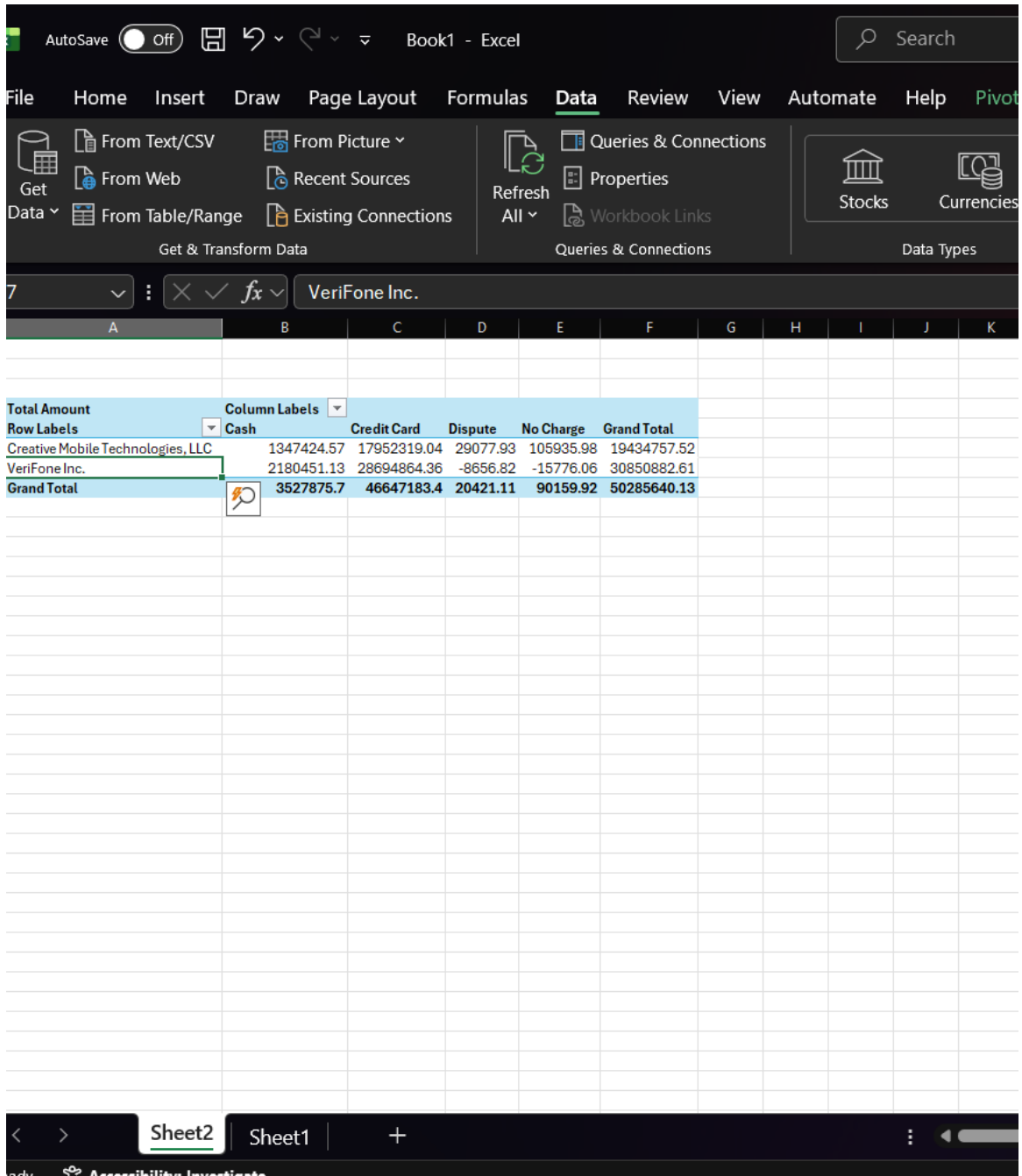
The PivotTable Fields task pane on the right shows the following configuration:

- Choose fields to add to report:** Search bar, Rate Code Key (unchecked), **Rate Code Name** (checked), **Dim Vendor** (checked), Vendor ID (unchecked), Vendor Key (unchecked), **Vendor Name** (checked).
- Drag fields between areas below:**
 - Filters:** Vendor Name, Rate Code Name.
 - Columns:** (Empty)
 - Rows:** Payment Name.
 - Values:** Total Amount.
- Defer Layout Update:** (unchecked)
- Update:** (button)

The status bar at the bottom indicates "Ready" and "Accessibility: Investigate". The zoom level is set to 70%.

3.6.Pivot

The pivot operation was demonstrated by changing the arrangement of dimensions between the row and column areas of the PivotTable. In the final view, Vendor Name was placed in the rows while Payment Name was placed in the columns, allowing the same data to be viewed from a different analytical perspective.



AutoSave Off Book1 - Excel

File Home Insert Draw Page Layout Formulas Data Review View Automate Help Pivot

Get Data From Text/CSV From Picture Recent Sources From Web From Table/Range Existing Connections Refresh All Queries & Connections Properties Workbook Links Stocks Currencies

7 VeriFone Inc.

Total Amount	Column Labels	Cash	Credit Card	Dispute	No Charge	Grand Total
Row Labels						
Creative Mobile Technologies, LLC		1347424.57	17952319.04	29077.93	105935.98	19434757.52
VeriFone Inc.		2180451.13	28694864.36	-8656.82	-15776.06	30850882.61
Grand Total		3527875.7	46647183.4	20421.11	90159.92	50285640.13

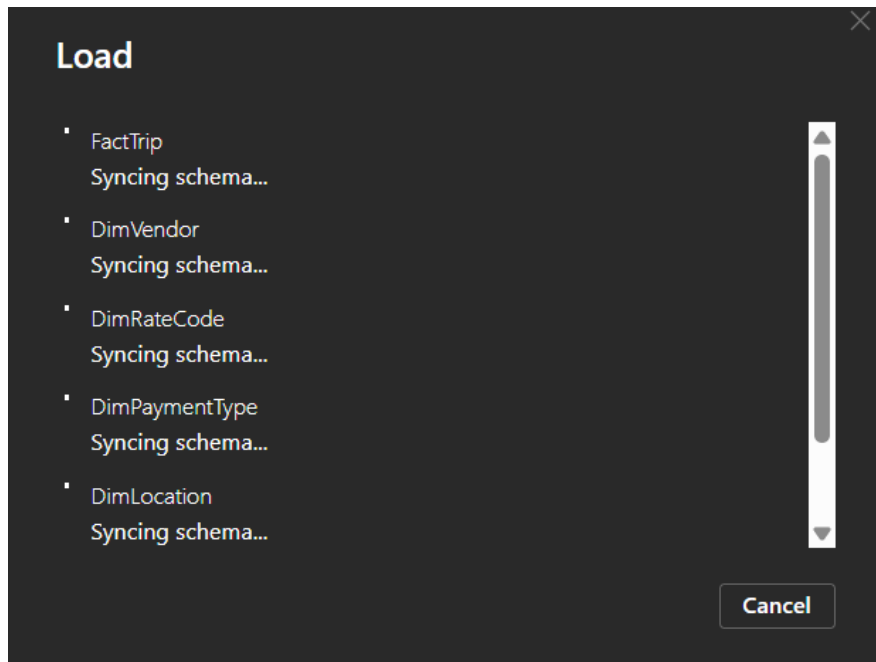
Sheet2 Sheet1

4. Power BI Reports.

4.1.Data Preparation and Modeling

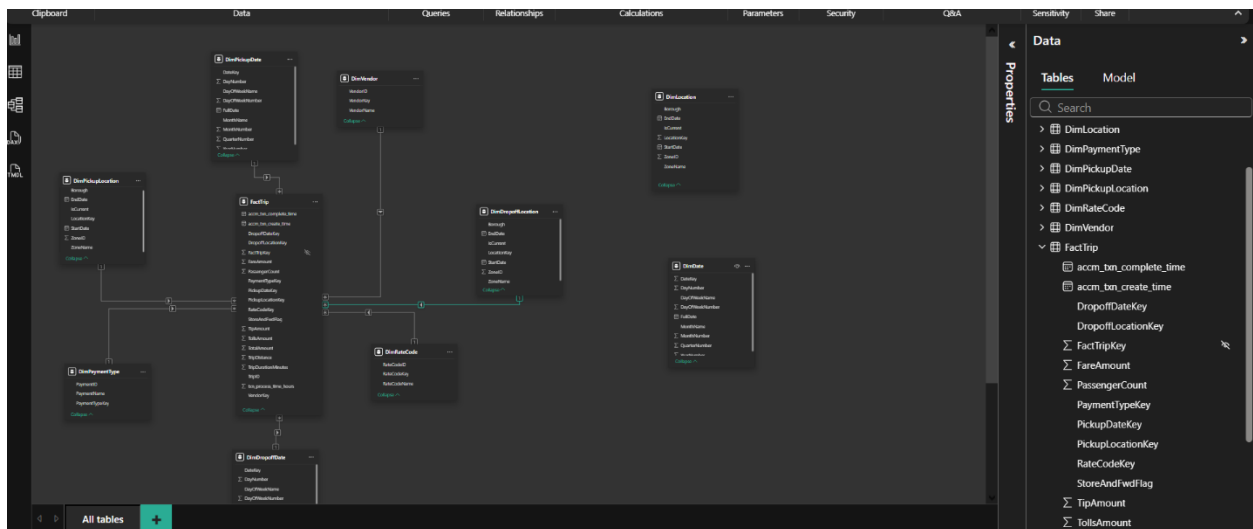
Power BI Desktop was used to create the required reports. The data was imported from the TAXI_TRIP_DATA_DW SQL Server database. The tables loaded were:

- FactTrip
- DimDate
- DimLocation
- DimVendor
- DimPaymentType
- DimRateCode



To support pickup and drop-off reporting clearly, duplicate dimension tables were created in Power BI:

- DimPickupDate
- DimDropoffDate
- DimPickupLocation
- DimDropoffLocation



Relationships were configured between the fact table and these dimensions. In addition, DAX measures were created for reporting purposes, including:

- Total Revenue
- Total Trips
- Total Distance
- Total Passengers
- Average Fare

```
1 Total Trips = COUNTROWS(FactTrip)
```

```
1 Total Revenue = SUM(FactTrip[TotalAmount])
```

```
1 Total Passengers = SUM(FactTrip[PassengerCount])
```

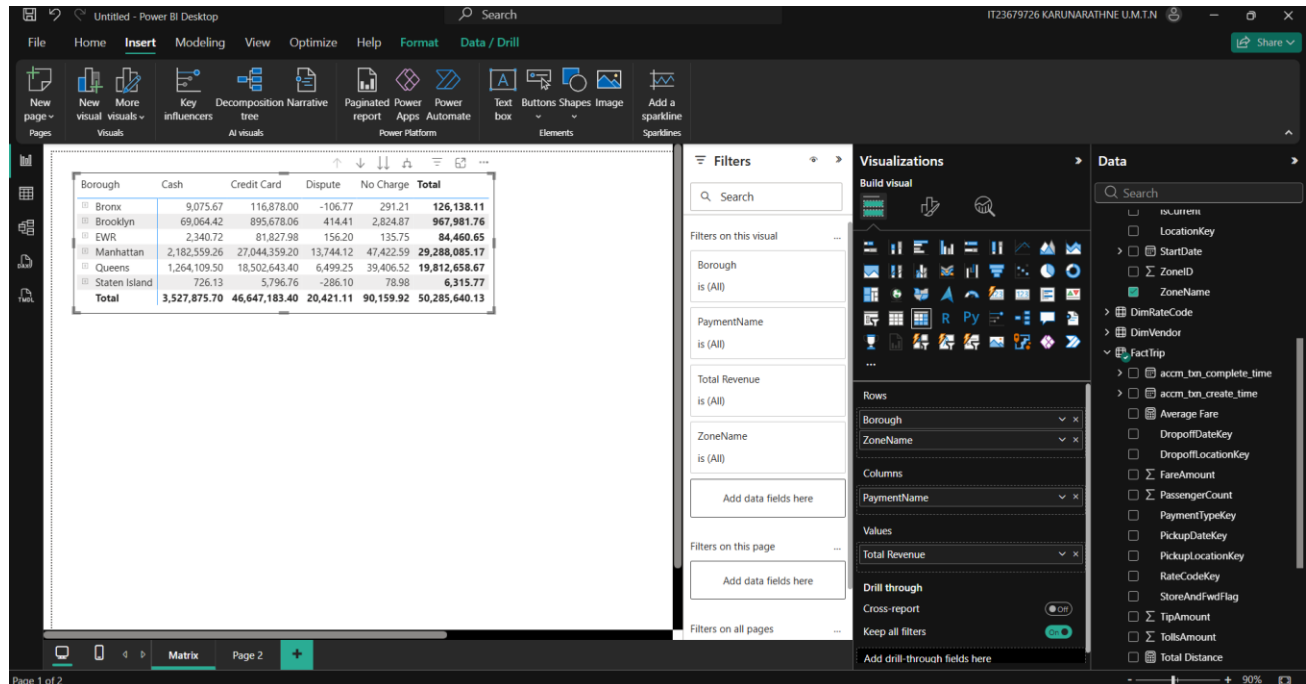
```
1 Total Distance = SUM(FactTrip[TripDistance])
```

```
1 Average Fare = AVERAGE(FactTrip[FareAmount])
```

4.2. Report 1 – Matrix Visual

The report used Borough and ZoneName from the Pickup Location dimension as row groups, PaymentName as the column group, and Total Revenue as the measure.

This report provides a compact analytical summary of revenue distribution across locations and payment methods.

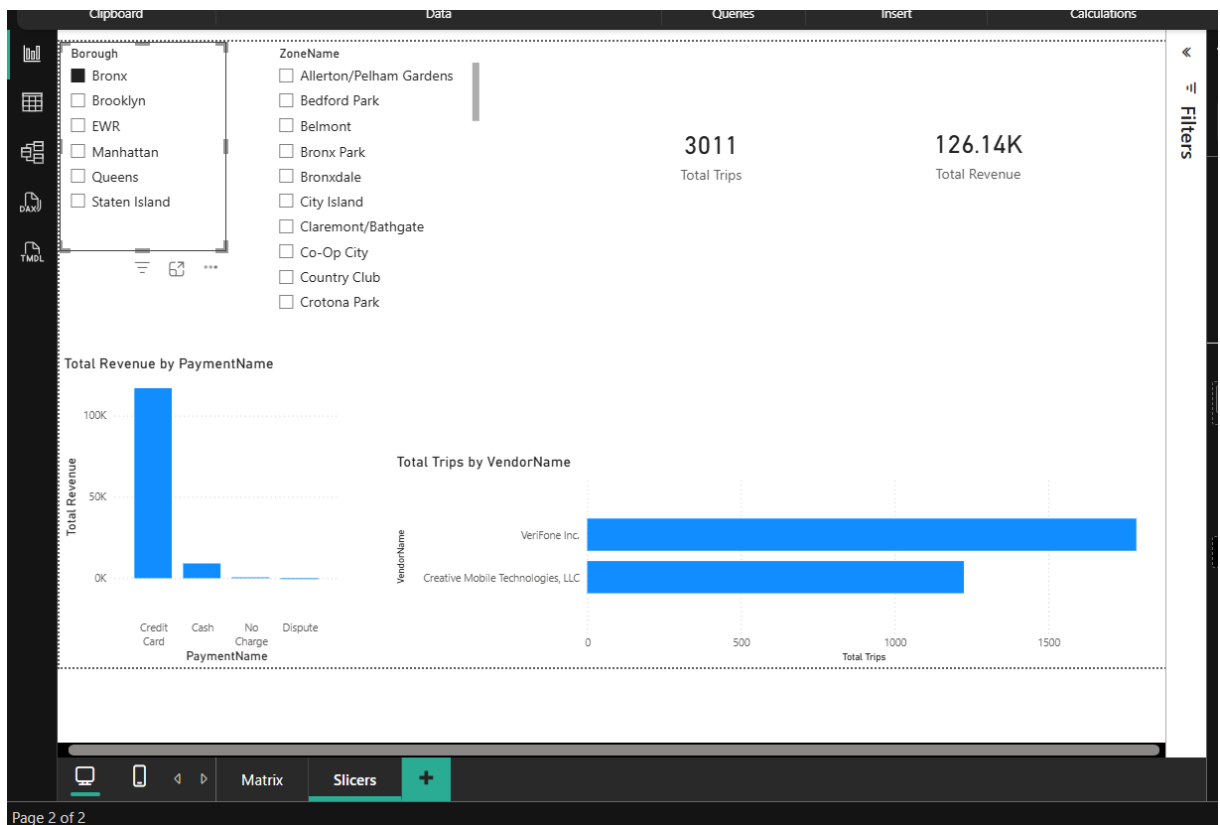
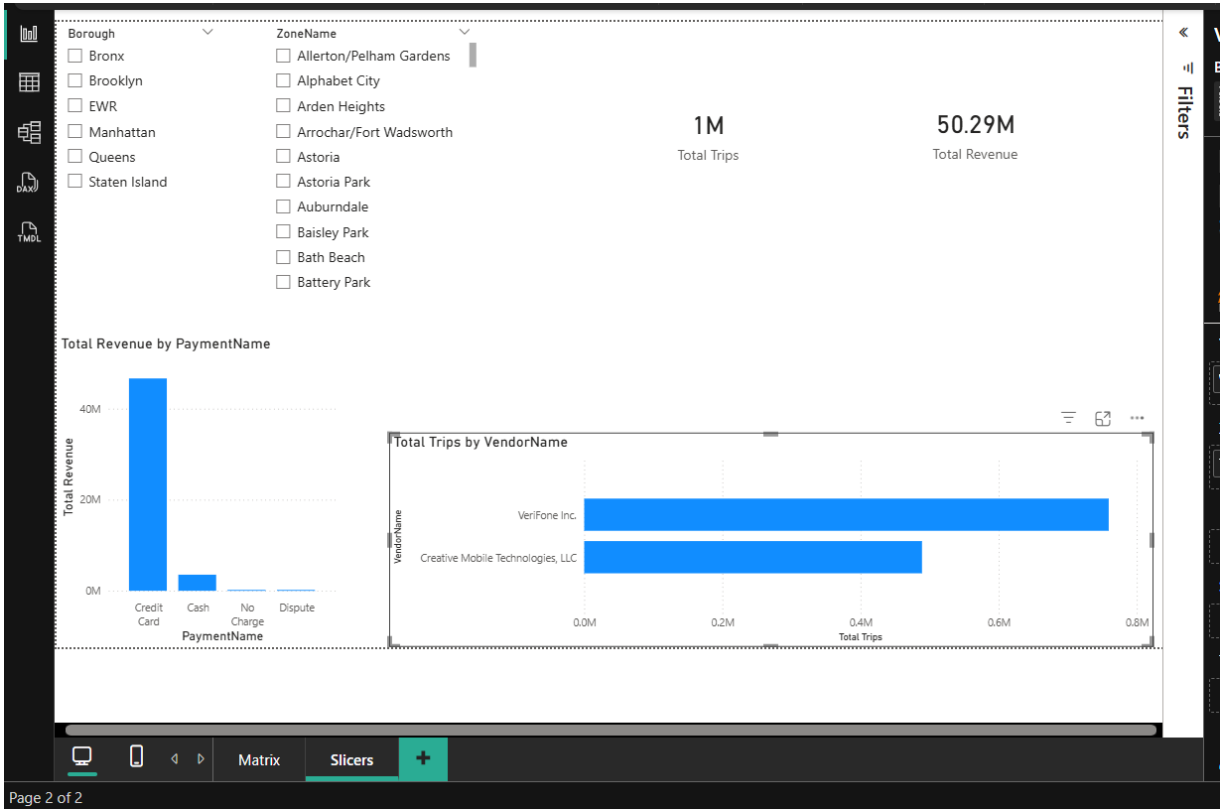


4.3. Multiple Slicers with Cascading Filtering

The first slicer used Borough and the second slicer used ZoneName from the Pickup Location dimension. When a Borough was selected, the ZoneName slicer displayed only the zones related to that borough, demonstrating cascading filtering.

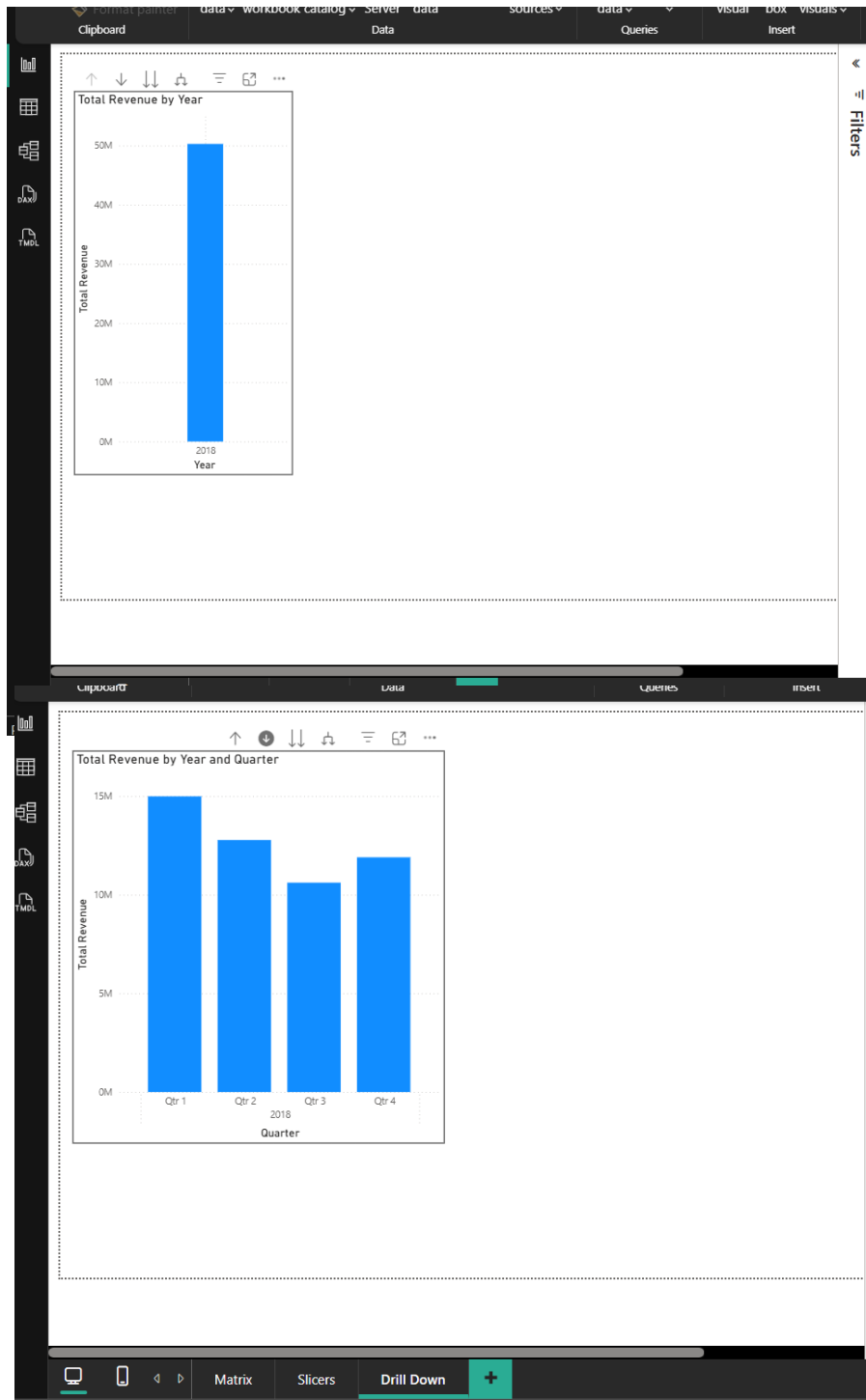
The report also included:

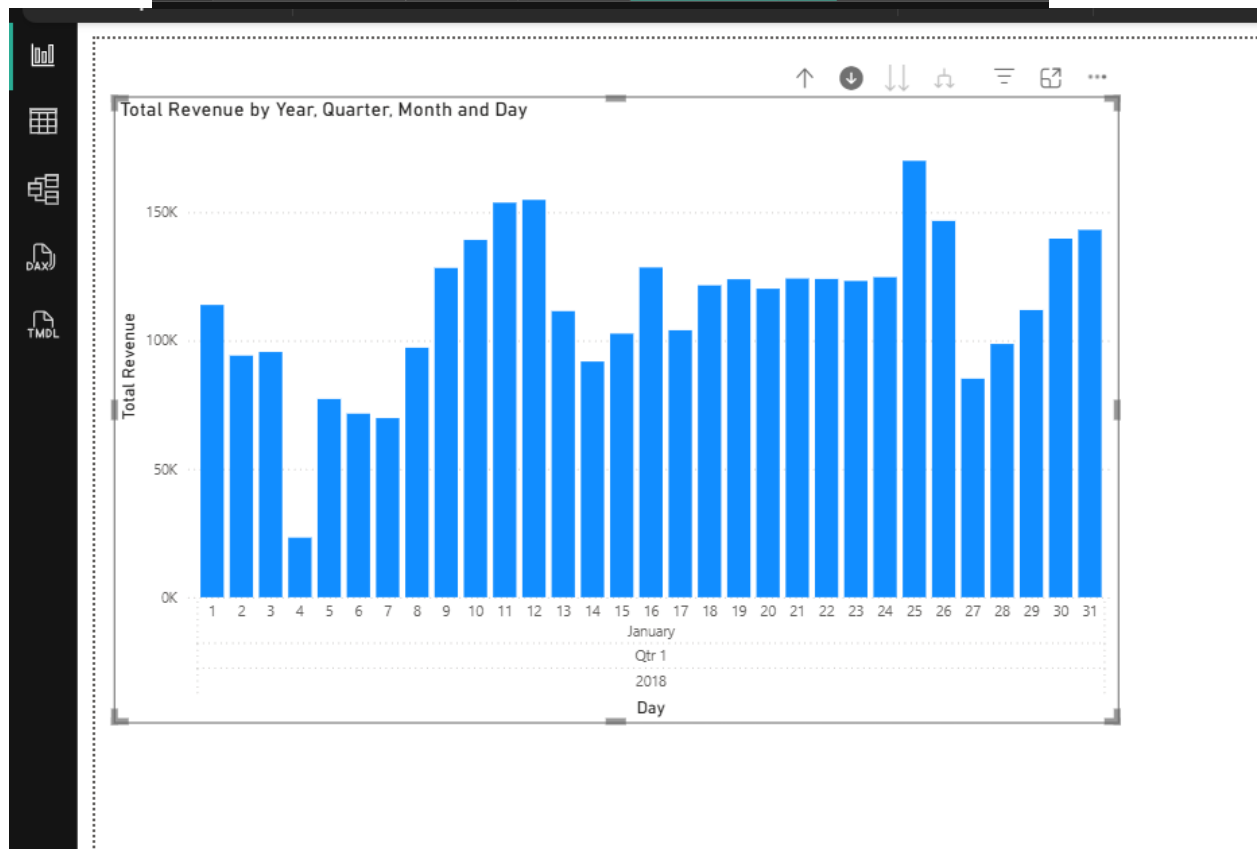
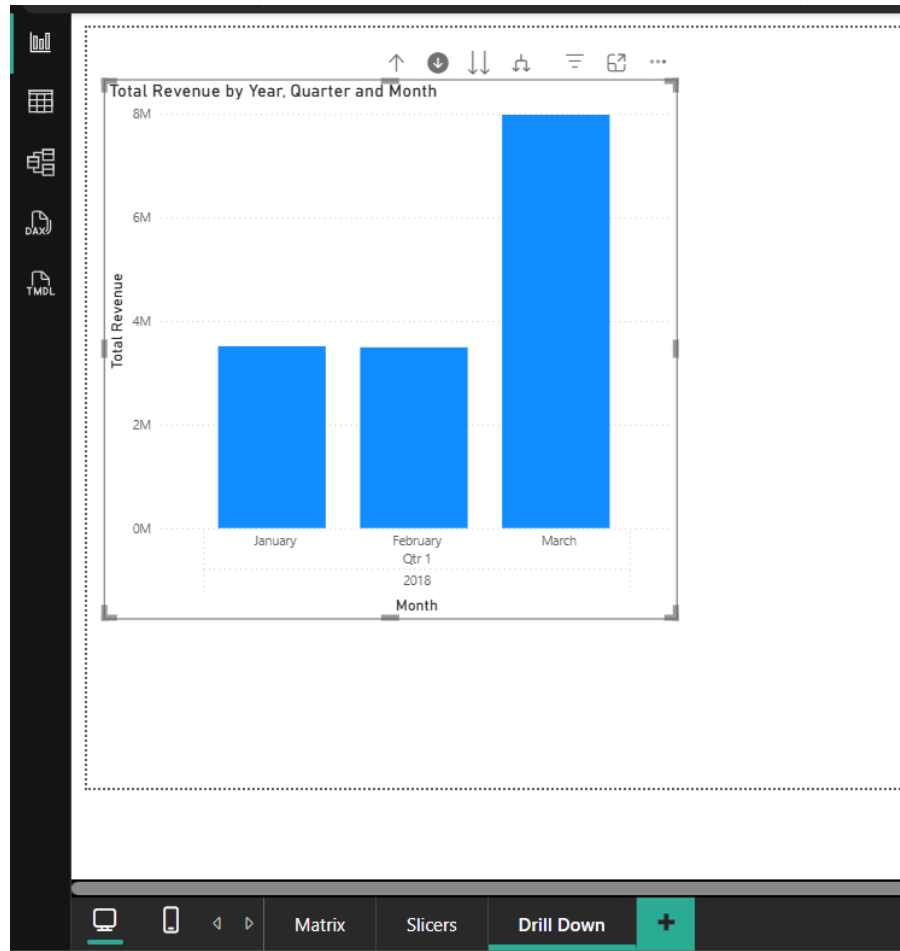
- a column chart for Total Revenue by Payment Type
- a bar chart for Total Trips by Vendor
- card visuals for Total Trips and Total Revenue



4.4. Report 3 – Drill-Down Report

A drill-down report was created in Power BI using the Date Hierarchy available in the Pickup Date dimension. A clustered column chart was used to display Total Revenue across time. The visual initially displayed yearly values and was then drilled down to lower levels such as quarter, month, and day.



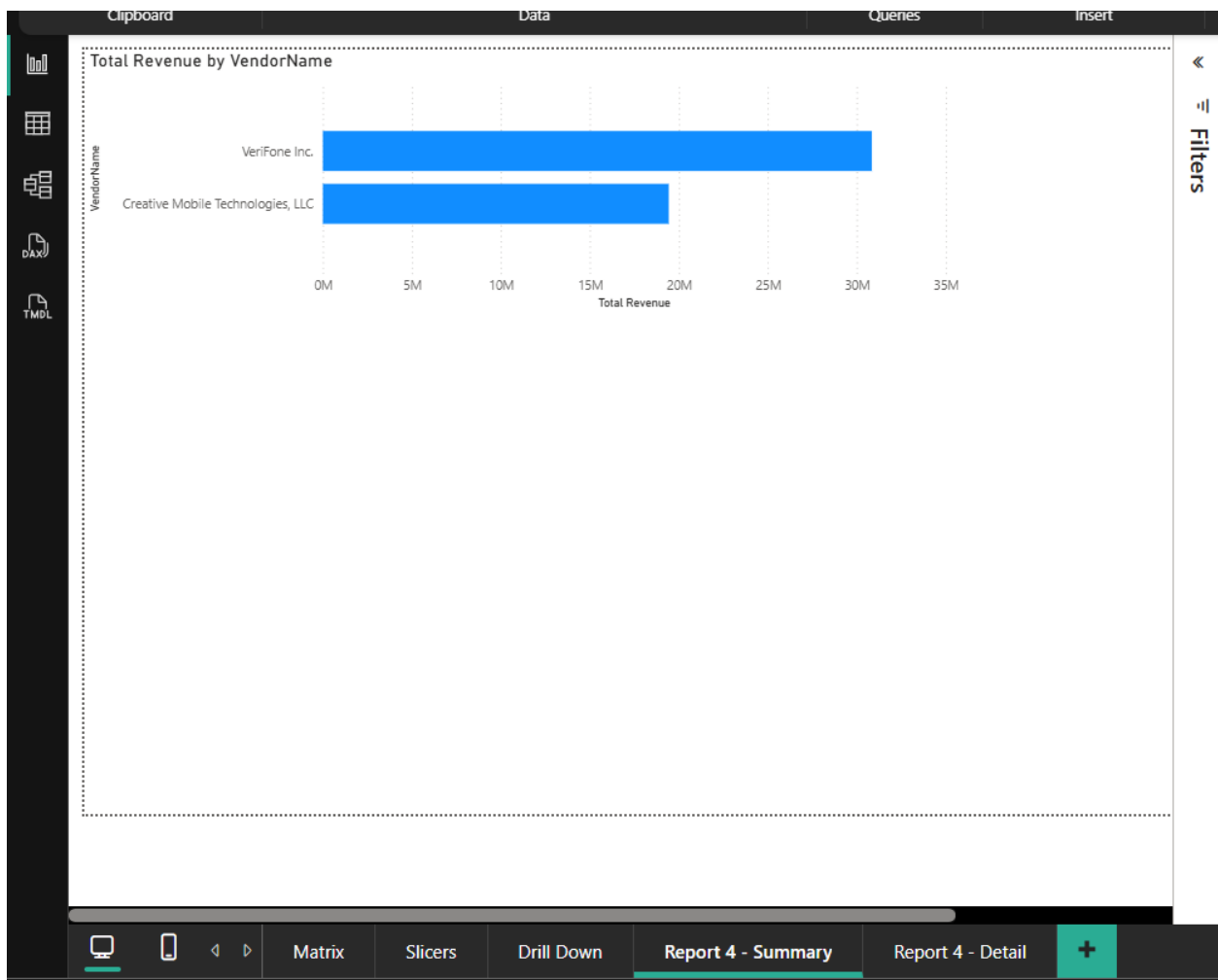


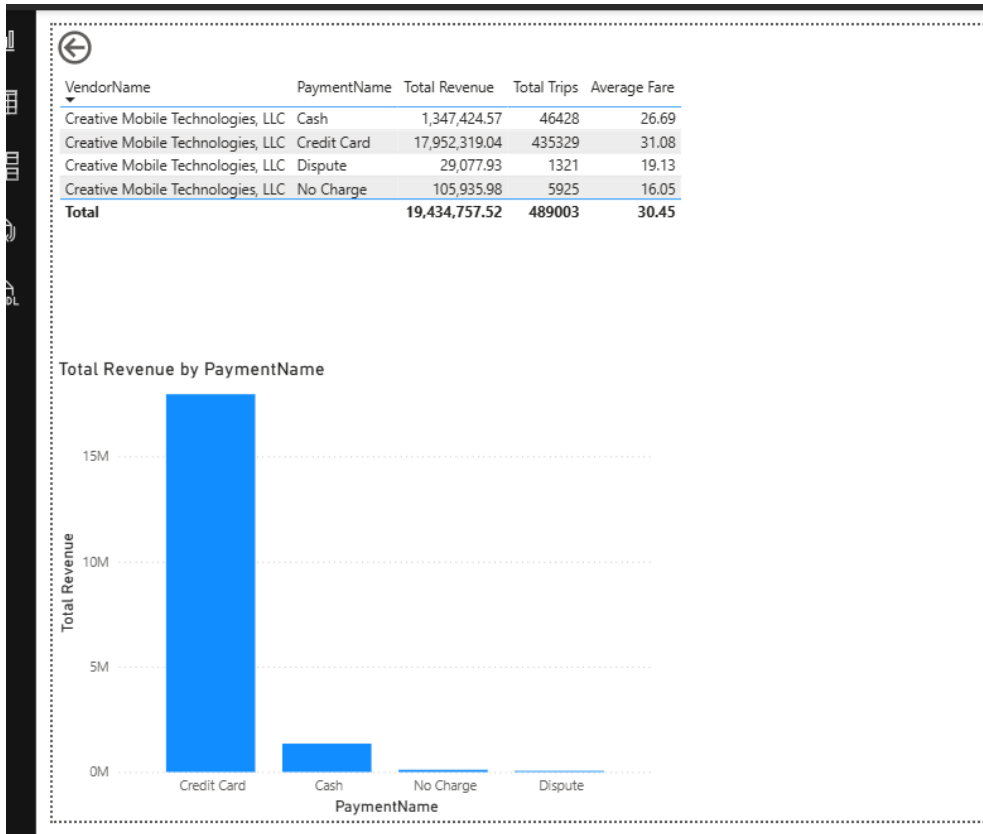
4.5. Report 4 – Drill-Through Report

A drill-through report was created to enable navigation from a summary page to a detailed page. The summary page displayed Total Revenue by Vendor using a bar chart. A separate detail page was configured with VendorName as the drill-through field.

When a user right-clicked a vendor in the summary report and selected the drill-through option, Power BI opened the detail page filtered to the selected vendor. The detail page included:

- a table visual with VendorName, PaymentName, Total Revenue, Total Trips, and Average Fare
- a chart showing Total Revenue by PaymentName
- a back button for navigation to the summary page





4.6.Publish to Power BI Service

After creating the reports in Power BI Desktop, the .pbix file was published to Power BI Service. The report was then opened online to confirm that the published version was accessible and functioning correctly.

